

ZHIBO YU

Department of Astronomy and Astrophysics
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EDUCATION

Pennsylvania State University

Graduate Fellow in PhD Program in Astronomy and Astrophysics
Verne M. Willaman Distinguished Graduate Fellowship in Science
Advisor: Professor Niel Brandt

University Park, PA, US

Aug. 2022 – present

Fudan University

Bachelor of Science in Physics (Honored Degree)
Advisor: Professor Cosimo Bambi

Shanghai, China

Sept. 2018 – June 2022

University of California, Berkeley

Berkeley Global Access Program

Berkeley, CA, US

Jan. 2020 – May 2020

PRIMARY RESEARCH INTERESTS

- Deep and wide cosmic multiwavelength surveys (LSST, Roman, Euclid, etc.)
- Astronomical Instrumentation
- Sample-based studies of active galactic nuclei
- The growth of supermassive black holes over cosmic time
- Black-hole vs. galaxy connections
- Statistical methods and machine-learning
- High-energy astrophysics and X-ray astronomy

PUBLICATIONS

First/Leading author (* indicates student mentees as first authors)

1. Tee, W. L.; Brandt, W. N.; **Yu, Z.**; Zou, F.; Zhang, B.; “Deep Multiwavelength Constraints on the Host Galaxies and Black-Hole Growth of Dwarf AGNs”. Submitted to ApJ
2. **Yu, Z.**; Brandt, W. N.; Zou, F.; Luo, B.; Ni, Q.; Schneider, D. P.; Vito, F. “The Drivers of the Decline in Supermassive Black Hole Growth at $z < 2$ ”. ApJ, 995(2), 205 (2025)
3. *Sullivan, Q. P.; **Yu, Z.**; Brandt, W. N.; Luo, B.; Zou, F. “The Nature of an Unidentified X-ray Source in the 7Ms Chandra Deep Field-South Catalog”. RNAAS, 9, 204 (2025)
4. **Yu, Z.**; Brandt, W. N.; Zou, F.; Zhu, Z.; Bauer, F.; Ni, Q.; Vito, F.; Luo, B.; Xue, Y. “Dust-Obscured Galaxies in the XMM-SERVS Fields: Selection, Multiwavelength Characterization, and Physical Nature”. ApJ, 977(2), 210 (2024)
5. Cristello, N.; Zou, F.; Brandt, W. N.; **Yu, Z.**; Vito, F.; Zhu, S. “A Rapidly Accreting Active Galactic Nucleus Hidden in a Dust-Obscured Galaxy at $z \sim 0.8$ ”. ApJ, 977(1), 113 (2024)
6. Zou, F.; **Yu, Z.**; Brandt, W. N.; Tak, H.; Yang, G.; Ni, Q. “Mapping the Growth of Supermassive Black Holes as a Function of Galaxy Stellar Mass and Redshift”. ApJ, 964(2), 183 (2024)
7. *Wang, S.; Brandt, W. N.; Luo, B.; **Yu, Z.**; Zou, F.; Huang, J.; Ni, Q.; Vito, F. “The Remarkable X-ray Spectra and Variability of the Ultraluminous Weak-Line Quasar SDSS J1521+5202”. ApJ, 974(1), 2 (2024)
8. **Yu, Z.**; Zou, F.; Brandt, W. N. “Stellar Masses and Star-Formation Rates of Galaxies and AGNs in the eFEDS GAMA09 Field”. RNAAS, 7, 248 (2023)
9. **Yu, Z.**; Jiang, J.; Bambi, C.; Gallo, L. C.; Grupe, D.; Fabian, A. C.; Reynolds, C. S.; Brandt, W. N. “An XMM-Newton Study of Six Narrow-Line Seyfert 1 Galaxies at $z = 0.35-0.92$ ”. MNRAS, 522, 5456 (2023)
10. **Yu, Z.**; Jiang, Q.; Abdikamalov, A. B.; Ayzenberg, D.; Bambi, C.; Liu, H.; Nampalliwar, S.; Tripathi, A. “Constraining the Konoplya-Rezzolla-Zhidenko deformation parameters. II. Limits from stellar-mass black hole X-ray data”, Phys. Rev. D, 104, 084035 (2021)

Contributing author:

11. Satheesh-Sheeba, S.; Sánchez-Sáez, P.; Assef, R. J. and others, including **Yu, Z.** “VAR-PZnn: A Machine-Learning Framework for AGN Photometric Redshifts using Color and Variability-Based Features”. Submitted to A&A
12. Zou, F.; Gallo, E.; Tee, W. L.; Brandt, W. N.; Nguyen, D. D.; Tang, S.; **Yu, Z.** “Mapping the Growth of Massive Black Holes as a Function of Galaxy Stellar Mass and Redshift II: Late-Time Accretion Dominates the Black Hole Budget in Low-Mass Galaxies”. Submitted to ApJ

13. Shirley, R.; Salvato, M.; Cohen-Tanugi, J. and others, including **Yu, Z.** “Photometric Redshifts for Active Galactic Nuclei with LePHARE for the Vera C. Rubin Observatory”. Accepted by A&A.
14. Zou, F.; Brandt, W. N.; Gallo, E.; Vito, F.; **Yu, Z.** “The X-ray Link Between High Eddington Ratio Dust-Obscured Galaxies (DOGs) and Hot DOGs”. *ApJ*, 990(9), 4 (2025)
15. Jiang, J.; Walton, D. J.; Gallo, L. C.; Fabian, A. C.; Grupe, D.; McMahon, R.; Reynolds, C. S.; Young, A.; **Yu, Z.**; Liu, H.; Zhang, Z. “Narrow-Line Seyfert 1 Galaxies Beyond the Local X-ray Universe: An X-ray Spectral Sample”. *MNRAS*, 539, 832 (2025)
16. Zhang, B.; Zou, F.; Brandt, W. N.; Zhu, S.; Cristello, N.; Ni, Q.; Xue, Y.; **Yu, Z.** “Investigating the Star-Formation Characteristics of Radio Active Galactic Nuclei”. *ApJ*, 978(1), 18 (2025)
17. Zou, F.; Brandt, W. N.; Gallo, E.; Luo, B.; Ni, Q.; Xue, Y.; **Yu, Z.** “The Cosmic Evolution of the Supermassive Black Hole Population: A Hybrid Observed Accretion and Simulated Mergers Approach”. *ApJ*, 976(1), 6 (2024)
18. Zhang, Z.; Jiang, J.; Liu, H.; Bambi, C.; Reynolds, C. S.; Fabian, A. C.; Dauser, T.; Madsen, K.; Young, A.; Gallo, L.; **Yu, Z.**; Tomsick, J. “The Low-temperature Corona in ESO 511-G030 Revealed by *NuSTAR* and *XMM-Newton*”. *ApJ*, 949(1), 4 (2023)
19. Riaz, S.; Abdikamalov, A.; Ayzenberg, D.; Bambi, C.; Wang, H.; **Yu, Z.** “Reflection spectra of accretion disks illuminated by disk-like coronae”, *ApJ*, 925(1), 51 (2022)

APPROVED PROPOSALS AND GRANTS

1. Roman Space Telescope Cycle 1, Analysis Proposal (19007; **\$150K**), **Co-I**, “Roman and the Rise of Giants: Black-Hole Growth in Forming Galaxy Clusters”.
2. Hobby-Eberly Telescope PSU26-1-008, 1.78 hrs, **P.I.**, “Testing the Merger-Driven Coevolution Framework for a Dwarf AGN”.
3. Hobby-Eberly Telescope PSU26-1-009 & PSU26-2-009, total 3.92 hrs, **P.I.**, “Identifying Coronal Lines for the Smallest Massive Black Holes”.
4. Chandra X-ray Observatory Cycle 25, Archive Proposal (25700139; **\$75K**), **P.I.**, “Understanding the Black-Hole Accretion - Stellar-Mass Relation Over All of Cosmic Time”.

SELECTED PRESS RELEASE

1. “NASA Resolves Why Black Holes Hit the Brakes on Growth”
CXC: https://chandra.si.edu/press/26_releases/press_032426.html
Penn State: <https://www.psu.edu/news/eberly-college-science/story/supermassive-black-holes-are-growing-slower-because-they-have-less>
U-Michigan: <https://news.umich.edu/why-black-holes-hit-the-brakes-on-growth/>

SELECTED AWARDS AND HONORS

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|---|------------------------|
| • Penn State International Travel Award (\$1500) | Penn State, Mar. 2026 |
| • Larry Travis Fellowship (\$1100) | Penn State, May 2025 |
| • Edward M. Frymoyer Award (\$3900) | Penn State, Nov. 2024 |
| • Sampson Memorial Award (\$3000) | Penn State, Apr. 2024 |
| • Zaccheus Daniel Fellowship (~\$800) | Penn State, 2023-2024 |
| • Verne M. Willaman Distinguished Graduate Fellowship in Science (~\$45K) | Penn State, Sept. 2022 |
| • Representative of Undergraduate Students at 2022 Fudan Commencement Ceremony | Fudan, June 2022 |
| • Undergraduate Research Scholarship (only for top five undergraduate researchers at Fudan) | Fudan, Nov. 2021 |
| • Bao-Steel Scholarship (1/88 in Department of Physics; only for top four students at Fudan) | Fudan, Sept. 2021 |
| • Summer Undergraduate Research Fellowship (SURF) at Caltech (\$6600) | Caltech, Apr. 2021 |
| • Championship of China Undergraduate Physics Tournament (record-breaking for Fudan: ranked top one for the first time; a team-oriented physics competition between sixty top universities in China) | Aug. 2019 |

CONFERENCES, WORKSHOPS, AND INVITED TALKS

- LSST AGN Science Collaboration Meeting 2026**, Heraklion, Crete, Greece May 25-28th, 2026
- Talk: The Drivers of the Decline in Supermassive Black Hole Growth at $z < 2$.
- 2025 APS Mid-Atlantic Section**, Penn State Harrisburg, PA, US Nov. 14-16th, 2025
- Talk: The Drivers of the Decline in Supermassive Black Hole Growth at $z < 2$.
- 2025 Central Pennsylvania Consortium Astronomers' Meeting**, Dickinson College, PA, US Apr. 5th, 2025
- Talk: Dust-Obscured Galaxies in the XMM-SERVS Fields: Selection, Multiwavelength Characterization, and Physical Nature
- Unveiling a Universe of Black Holes: the Next Generation of AGN Surveys**, Garching, Germany Mar. 10-21th, 2025

- Talk: Dust-Obscured Galaxies in the XMM-SERVS Fields: Selection, Multiwavelength Characterization, and Physical Nature
- (Invited) The Extreme Astrophysics Group Meeting at University of Michigan, virtual** Feb. 19th, 2025
- Talk: Dust-Obscured Galaxies in the XMM-SERVS Fields: Selection, Multiwavelength Characterization, and Physical Nature
- 25 Years of Science with Chandra, Boston, MA, US** Dec. 3–6th, 2024
- Poster: The Remarkable X-ray Spectra and Variability of the Ultraluminous Weak-Line Quasar SDSS J1521+5202
- The 21st meeting of the High Energy Astrophysics Division of the AAS, Horseshoe Bay, TX, US** Apr. 7–12th, 2024
- Poster: Stellar Masses and Star-Formation Rates of Galaxies and AGNs in the eFEDS GAMA09 Field
- New Era of AGN Science with the Vera C. Rubin LSST, Charlottesville, VA, US** Jul. 24–26th, 2023
- Neighborhood Workshop on Astrophysics and Cosmology, University Park, PA, US** 2023–2026
- 2022 APS Mid-Atlantic Section, University Park, PA, US** Dec. 3–5th, 2022
- Talk: An XMM-Newton Study of Narrow-Line Seyfert 1 Galaxies at $z = 0.35-0.92$
- The 9th Microquasar Workshop, virtual** Sept. 20–24th, 2021
- Poster: Broad-band spectral-timing analysis on MAXI J1820+070 with Insight-HXMT data
- Timing X-ray Binaries: On the road to eXTP, virtual** Oct. 7–8th, 2021
- China-India Workshop on High Energy Astrophysics, virtual** Nov. 6–7th, 2020
- Accretion 2020 @ Fudan, virtual** Oct. 21–23rd, 2020

SERVICE AND OUTREACH

- Guest lecture, Astro 120: The Big Bang Universe 2024
- Volunteer, Astro Night, Department of Astronomy and Astrophysics at Penn State 2022–2025
- Volunteer, Astro Fest, Department of Astronomy and Astrophysics at Penn State 2023–2026
- Special Correspondent for popular science on Fudan Astronomy Society journal 2019–2022
- Representative and event organizer for the Fudan Astronomy Society (FAS) 2019–2022
- Leader of the Literature and Art Department, Students' Union in the Department of Physics 2019–2020
- Speaker, FAS, *How to Begin Night Sky and Star Photography* Oct. 2021
- Volunteer, East Asia Regional Office of Astronomy for Development of IAU Aug. 2021

ADDITIONAL INFORMATION

Computer and Language Skills

- Languages: Python, Julia, C/C++
- Softwares: Mathematica, Matlab, CORSIKA, Arduino, Solidworks, COMSOL

SED-fitting techniques

- CIGALE

X-ray analysis techniques

- Xspec, PyXspec, Stingray, PIMMS
- Familiar with instruments: *XMM-Newton*, *Chandra*, *NuSTAR*, *Insight-HXMT*